

Rates and Predictors of Suicidal Ideation During the First Year After Traumatic Brain Injury

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Suicide is a major public health problem among the 1.7 million people who sustain traumatic brain injury (TBI) each year in the United States.¹ People with a history of TBI in both civilian and military populations are 1.55 to 4.05 times more likely to die by suicide than the general population.^{2–5} In a study of Australian outpatients with a history of TBI, the majority of whom had no preinjury history of suicide attempts, suicide attempts were reported by 17.4% (30 of 172) of the sample over a 5-year period.⁶ Nearly half of the individuals who attempted suicide had made multiple attempts.^{6,7} The Centers for Disease Control and Prevention recently called for investigations of individual-level risk and protective factors for self-directed violence among people with TBI as an important component of improving long-term outcomes.⁸

Rates of suicidal ideation (SI) after TBI have been found to exceed 20% in some studies^{6,9–14}; however, in a recent systematic review of SI and behavior after TBI, Bahraini et al. highlighted the paucity of research in this area.¹⁵ They concluded that additional research is needed to determine the prevalence of SI and behavior after brain injury, as well as to ascertain patient-level factors that may be associated with increased suicide risk. Studies examining whether injury severity predicts post-TBI suicidality have yielded inconclusive findings.^{6,13,16,17} In perhaps the most thorough study on this topic to date, Tsousides et al.¹² surveyed 356 community-dwelling adults with a self-reported history of TBI and found that preinjury substance abuse was the only correlate of current SI. Risk factors for SI after TBI have been underinvestigated. Research in this area has been limited by reliance on retrospective reporting and self-reported history of TBI,^{12,18–20} with only a few studies including objective indicators of TBI severity.⁶ Most studies have involved cross-sectional designs and have included participants whose time

Objectives. We examined rates of suicidal ideation (SI) after traumatic brain injury (TBI) and investigated whether demographic characteristics, preinjury psychiatric history, or injury-related factors predicted SI during the first year after injury.

Methods. We followed a cohort of 559 adult patients who were admitted to Harborview Medical Center in Seattle, Washington, with a complicated mild to severe TBI between June 2001 and March 2005. Participants completed structured telephone interviews during months 1 through 6, 8, 10, and 12 after injury. We assessed SI using item 9 of the Patient Health Questionnaire (PHQ-9).

Results. Twenty-five percent of the sample reported SI during 1 or more assessment points. The strongest predictor of SI was the first PHQ-8 score (i.e., PHQ-9 with item 9 excluded) after injury. Other significant multivariate predictors included a history of a prior suicide attempt, a history of bipolar disorder, and having less than a high school education.

Conclusions. Rates of SI among individuals who have sustained a TBI exceed those found among the general population. Increased knowledge of risk factors for SI may assist health care providers in identifying patients who may be vulnerable to SI after TBI. (*Am J Public Health.* 2014;104:e100–e107. doi:10.2105/AJPH.2013.301794)

since injury varied from several months to many years.^{12,21} Finally, because most existing studies have included relatively small, potentially biased samples²¹ recruited from outpatient clinics or TBI survivor programs,^{6,7,12} they may not be representative of the population of people who sustain TBI.

Given these gaps in the existing literature, our objectives were (1) to investigate rates of SI during the first year after complicated mild to severe TBI in a representative sample of adults who had been admitted to a level I trauma center and (2) to investigate whether demographic characteristics, preinjury psychiatric history, or injury-related factors predicted SI.

METHODS

This study was part of the recruitment phase of a clinical trial investigating the efficacy of the antidepressant sertraline for major depressive disorder after TBI.²² Eligibility criteria were as follows:

1. admitted to Harborview Medical Center, a level 1 trauma center in Seattle,

Washington, between June 2001 and March 2005;

2. at least 18 years old;
3. English speaking;
4. a resident of King, Pierce, Kitsap, Jefferson, Mason, Thurston, or Snohomish counties; and
5. radiological or clinical evidence of acute TBI (i.e., Glasgow Coma Scale [GCS] score < 13 within the first 24 hours after admission, provided blood alcohol level was < 200 mg/dL).

For participants whose GCS score at the time of admission was 13 to 15, radiological evidence was necessary for categorization of complicated mild TBI. Participants with uncomplicated mild TBIs (GCS score of 13–15 without radiological abnormality) were excluded. Other exclusion criteria were current homelessness, incarceration, lack of contact information, or diagnosis of schizophrenia.

The study team reviewed automatic queries of electronic medical records and TBI consultation lists on a daily basis to identify eligible inpatients. Research staff obtained consent

from potential participants, who were fully oriented during their admission prior to discharge. Legal next of kin provided assent to follow-up for patients who were disoriented at discharge. Patients who were unable to be approached during the course of their hospital stay were contacted after discharge by telephone or via a letter from the attending neurosurgeon. We followed prospective participants who were not oriented at discharge for 1 year postinjury or until they were oriented and able to consent to participate or to decline. Prior to consent, all patients completed a standardized orientation test.²³

Measures

Baseline assessment. The study team extracted medical, radiological, and other injury-related data (e.g., toxicology reports) from patients' electronic medical records and the Harborview Trauma Registry. The intentionality of the TBI (i.e., unintentional, self-inflicted, inflicted by someone else) was documented for 98% of the sample. Demographic information obtained from those sources was supplemented by participant interviews, conducted via telephone by trained research staff. The initial assessment consisted of a structured interview to assess preinjury psychiatric and substance use history, as well as prior mental health treatment. We obtained participants' history of suicidal behavior, including history of suicide attempts, number of attempts, date of most recent attempt, and age at first attempt. We assessed lifetime history of psychiatric disorders by inquiring whether participants had ever been diagnosed with or treated for any of the following conditions: clinical depression or major depression; bipolar disorder or manic depression; posttraumatic stress disorder; or generalized anxiety disorder, obsessive compulsive disorder, panic disorder, or any phobia. We used the CAGE (cut down, annoyed, guilty, eye opener) screening tool²⁴ to assess lifetime history of alcohol dependence. We defined alcohol intoxication as a blood alcohol level greater than 79 milligrams per deciliter, and drug abuse as having a positive toxicology screen (available on 80% of the sample) for cocaine or amphetamine upon admission.

Follow-up assessments. The study team conducted structured telephone interviews

monthly from months 1 through 6 and bi-monthly during months 8, 10, and 12. Follow-up telephone interviews assessed SI; symptoms of depression, anxiety, mental health treatment; and other aspects of adjustment after TBI (e.g., return to employment). We designed the baseline and follow-up interview protocols for the purpose of this study, although some components were derived from previously validated measures. For instance, we administered at each time point the Patient Health Questionnaire (PHQ-9),²⁵ a self-report measure that queries how often in the last 2 weeks respondents have been bothered by the 9 core symptoms of major depressive disorder (per the *Diagnostic and Statistical Manual of Mental Disorders, Fourth Edition, Text Revision*²⁶). The PHQ-9 has demonstrated good psychometric properties among individuals with TBI.²⁷ We used an abbreviated version of the PHQ-9, the PHQ-8 (which excludes item 9), to track depressive symptoms independent of SI.^{28,29} We monitored SI using item 9 of the PHQ-9 ("Thoughts that you would be better off dead, or of hurting yourself"). The PHQ-9 asks about the preceding 2 weeks, and responses are rated on a 4-point Likert scale ("not at all," "several days," "more than half of days," "nearly every day"). We utilized additional modules from the PHQ to assess symptoms of anxiety.³⁰ When a participant reported SI with plan or intent at any assessment point, the study team contacted one of the investigators (C. H. B. or J. R. F.) immediately, and the participant was referred for further evaluation or treatment, as appropriate.

Statistical Analyses

We classified participants as SI positive if they reported SI (i.e., item 9 of PHQ-9 was > 0) at any assessment point and SI negative if they never reported SI. We estimated prevalence rates as the proportion of participants interviewed at a given assessment point who reported SI, regardless of whether it was the first time, a continuation, or a recurrence. We calculated cumulative postinjury prevalence as the proportion of participants who had ever reported SI since their injury up to that point. We calculated postinjury incidence as the proportion of participants who reported SI for the first time since their injury at a given assessment point. First, we used logistic regression to

examine bivariate relationships between SI and all demographic, preinjury, and injury-related variables of interest using LogXact version 4.1 (Cytel Software Corp, Cambridge, MA). We then implemented a forward stepwise regression procedure to identify independent predictors of SI during the study period. We created 2 models. The first model included variables generally available in a standard medical record review that were significant in the bivariate models. Race/ethnicity, marital status, and cause of injury were also included. The second model additionally considered all other variables that were significant in the bivariate models. Both models considered total number of interviews completed.

RESULTS

During the study period, we identified 1080 eligible patients, 559 of whom consented to participate in the study and completed at least 1 interview. Details about participants who were not enrolled are provided elsewhere.²² Participants were younger (mean age = 42.5 years [SD = 17.9] vs 46.8 years [SD = 21.5]), more likely to have completed high school (89% vs 84%), and less likely to have Medicare insurance (16% vs 25%) than patients who were not recruited into the study. Recruited versus nonrecruited groups did not differ with regard to gender, race/ethnicity, marital status, cause of injury, TBI severity, other injury severity, blood alcohol level, toxicology results, or length of hospital stay.

The recruited sample was composed predominantly of non-Hispanic White (86%) males (71.6%), and most had sustained their TBI as a result of a vehicular collision (41%; Tables 1 and 2). Approximately half (52%) of the brain injuries were categorized as complicated mild (i.e., GCS score of 13–15 with radiological evidence of acute TBI). Thirty percent of participants had a positive blood alcohol level on admission to the hospital. Toxicology screening indicated that 7% of the sample used cocaine and 6% used methamphetamine around the time of their injury. Twice as many SI-positive participants reported receiving mental health treatment at the time of injury (47% vs 23%). Fewer

TABLE 1—Demographic Characteristics of the Study Sample, by Presence or Absence of Postinjury Suicidal Ideation (SI): Seattle, WA, June 2001–March 2005

Variable	SI Negative (n = 421), No. (%)	SI Positive (n = 138), No. (%)	SI Positive, % ^a	OR (95% CI)
Age*				
18–29	133 (32)	42 (30)	24	2.31 (1.12, 5.10)
30–44	105 (25)	44 (32)	30	3.06 (1.48, 6.77)
45–59	95 (23)	40 (29)	30	3.07 (1.47, 6.87)
≥ 60 (Ref)	88 (21)	12 (9)	12	1.00
Gender				
Male (Ref)	310 (74)	90 (65)	23	1.00
Female	111 (26)	48 (35)	30	1.49 (0.96, 2.29)
Race/ethnicity				
Non-Hispanic White (Ref)	363 (87)	114 (84)	24	1.00
Hispanic	17 (4)	7 (5)	29	1.31 (0.45, 3.43)
Non-Hispanic African American	21 (5)	8 (6)	28	1.21 (0.45, 2.94)
Asian/Pacific Islander	12 (3)	5 (4)	29	1.33 (0.36, 4.15)
Other	6 (1)	2 (1)	25	1.06 (0.10, 6.04)
Education*				
< high school	34 (8)	24 (17)	41	2.38 (1.29, 4.33)
≥ high school (Ref)	385 (92)	114 (83)	23	1.00
Marital status				
Married (Ref)	164 (39)	47 (34)	22	1.00
Never married	175 (42)	54 (39)	24	1.08 (0.67, 1.72)
Divorced, separated, or widowed	82 (19)	37 (27)	31	1.57 (0.92, 2.69)
Preinjury employment status				
Working full-time (Ref)	172 (50)	71 (58)	29	1.00
Working part-time	56 (16)	12 (10)	18	0.52 (0.24, 1.06)
Unemployed	38 (11)	18 (15)	32	1.15 (0.58, 2.22)
Retired	54 (16)	7 (6)	11	0.32 (0.12, 0.74)
Other ^b	26 (8)	15 (12)	37	1.40 (0.65, 2.93)
Insurance*				
Commercial or private (Ref)	245 (58)	71 (51)	22	1.00
Medicaid	100 (24)	53 (38)	35	1.83 (1.17, 2.85)
Medicare	76 (18)	14 (10)	16	0.64 (0.31, 1.22)

Note. CI = confidence interval; OR = odds ratio.

^aPercentage with SI (response ≥ 1 on item 9 of the Patient Health Questionnaire [PHQ-9]) in each variable category.

^b“Other” includes medical leave of absence (n = 2), student (n = 11), homemaker (n = 1), and disabled (n = 24).

*P < .05; P values were determined by the Fisher exact test.

interviews took place during month 1 (n = 289) than at any other assessment point, as many participants had not yet become eligible (e.g., pending consent because they were still not oriented). Participants who were SI positive completed significantly fewer interviews than those who were SI negative (5.6 vs 6.1; $P = .04$), although participants continued to be interviewed regardless of their ideation status.

Rates of Suicidal Ideation and History of Suicidal Behavior

During the first year after injury, 25% of the sample reported SI at 1 or more assessment points. Point prevalence of SI ranged from 7% to 10% at each assessment point (Figure 1). Ideation was highest (10%) during months 2 and 8. Sixty percent of those who were SI positive screened positive during 1 interview only, whereas 22% reported SI at 3 or more

assessment points. Fifty-three percent of patients who reported suicidality during their first year after injury also reported probable depression (PHQ-8 score ≥ 10) at the time of their first assessment. According to their first PHQ-9 after injury, most SI-positive participants reported having thoughts that they would be better off dead or of hurting themselves on “several days” (37%), but a portion reported those feelings “more than half the days” (7%) or “nearly every day” (7%) in the prior 2 weeks.

Concerning prior history of suicidal behavior, 12% of the sample reported a history of 1 or more suicide attempts prior to their TBI (Table 3). The mean number of prior suicide attempts was similar for the SI-positive (1.7 [SD = 0.8]) and SI-negative (1.5 [SD = 0.6]) groups. Seven percent of participants who reported a prior suicide attempt indicated that they made 5 or more attempts. Relative to SI-negative participants, SI-positive participants reported being older the first time they attempted suicide, (median age = 28 years [interquartile range = 16–42] and 18 years [interquartile range = 15–23], respectively; $P = .04$). Those with a history of a suicide attempt were nearly 5 times more likely to report SI after injury than participants who denied a history of a suicide attempt (odds ratio = 4.81; 95% confidence interval = 2.83, 8.17). Eight individuals (1.4%) incurred their TBI as a result of self-inflicted injury. Of those, half reported SI during the study period.

Predictors of Suicidal Ideation

On the basis of bivariate relationships, the following factors were significantly associated with SI: having Medicaid insurance (relative to commercial or private); having a higher score on the first PHQ-8 after injury; a history of depression, bipolar disorder, or other anxiety disorder; and prior suicide attempt(s) or psychiatric hospitalization. Older age and retired status were associated with SI-negative status. There was a nonsignificant trend ($P = .07$) for females to be more likely to report SI. We created 2 stepwise regression models. In the first model, which included variables that are available in a standard medical record review, being female, having less than a high school education, and having Medicaid were significant predictors of SI in the year after injury

TABLE 2—Injury Characteristics of the Study Sample, by Presence or Absence of Postinjury Suicidal Ideation (SI): Seattle, WA, June 2001–March 2005

Variable	SI Negative (n = 421), No. (%)	SI Positive (n = 138), No. (%)	SI Positive, % ^a	OR (95% CI)
Cause of injury				
Vehicular (e.g., automobile, motorcycle; Ref)	172 (41)	56 (41)	25	1.00
Fall	144 (34)	40 (29)	22	0.85 (0.52, 1.39)
Violence	43 (10)	20 (15)	32	1.43 (0.73, 2.72)
Other or recreational	61 (15)	19 (14)	24	0.96 (0.50, 1.79)
Injuries were intentionally inflicted				
No (Ref)	367 (88)	112 (84)	23	1.00
Yes, self-inflicted	4 (1)	4 (3)	50	3.27 (0.60, 17.83)
Yes, inflicted by someone else	44 (11)	18 (13)	29	1.34 (0.70, 2.48)
GCS score				
Complicated mild (13–15; Ref) ^b	226 (54)	65 (47)	22	1.00
Moderate (9–12)	98 (23)	30 (22)	23	1.06 (0.63, 1.78)
Severe (3–8)	97 (23)	43 (31)	31	1.54 (0.95, 2.48)
Cortical contusions				
No (Ref)	294 (70)	101 (74)	26	1.00
Yes	125 (30)	36 (26)	22	0.84 (0.53, 1.32)
Intracerebral hemorrhage				
No (Ref)	126 (30)	48 (35)	28	1.00
Yes	293 (70)	89 (65)	23	0.80 (0.52, 1.23)
Injury score, ^c excluding head				
0 (Ref)	116 (28)	37 (27)	24	1.00
1–2	151 (36)	41 (30)	21	0.85 (0.50, 1.46)
3–5	152 (36)	60 (43)	28	1.24 (0.75, 2.06)
Length of hospital stay, d				
0–3 (Ref)	136 (32)	44 (32)	24	1.00
4–7	92 (22)	26 (19)	22	0.87 (0.48, 1.57)
8–14	99 (24)	28 (20)	22	0.87 (0.49, 1.55)
≥ 15	94 (22)	40 (29)	30	1.31 (0.77, 2.24)
Cocaine intoxication*				
No (Ref)	306 (93)	92 (84)	23	1.00
Yes	24 (7)	17 (16)	41	2.35 (1.13, 4.79)
Methamphetamine intoxication				
No (Ref)	307 (92)	108 (95)	26	1.00
Yes	26 (8)	6 (5)	19	0.66 (0.22, 1.69)
Alcohol level, mg/dL				
0 (Ref)	262 (68)	88 (68)	25	1.00
1–79	28 (7)	8 (6)	22	0.85 (0.32, 2.01)
≥ 80	97 (25)	34 (26)	26	1.04 (0.64, 1.69)
Litigation related to TBI*				
No (Ref)	267 (83)	80 (73)	23	1.00
Yes	55 (17)	29 (27)	35	1.76 (1.01, 3.02)
First PHQ-8 score after injury*				
< 10 (Ref)	324 (77)	65 (47)	17	1.00
≥ 10	97 (23)	73 (53)	43	3.74 (2.45, 5.73)

Note. CI = confidence interval; GCS = Glasgow Coma Scale; OR = odds ratio; PHQ-8 = Patient Health Questionnaire (PHQ-9) with item 9 excluded; TBI = traumatic brain injury.

^aPercentage with SI (response ≥ 1 on item 9 of PHQ-9) in each variable category.

^bParticipants with GCS scores of 13 to 15 had radiological evidence of TBI.

^cCalculated as the largest nonhead injury severity score.³¹

* $P < .05$; P values were determined by the Fisher exact test.

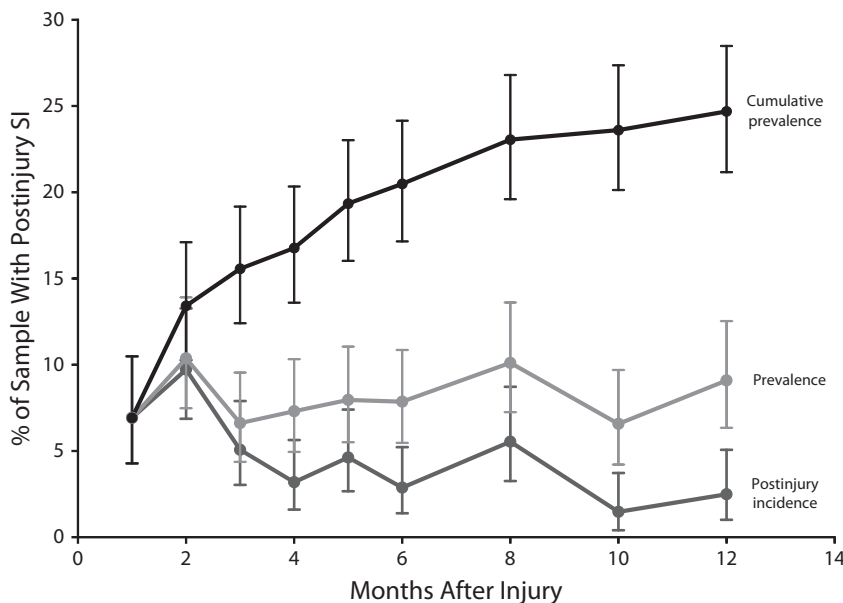


FIGURE 1—Prevalence, cumulative prevalence, and postinjury incidence of suicidal ideation (SI) at each assessment point during the first year after traumatic brain injury (n = 559): Seattle, WA, June 2001–March 2005.

(Table 4). In the overall model, the strongest predictor of SI during the year after TBI was the first PHQ-8 score after injury (Table 5). Other significant multivariate predictors included a history of a prior suicide attempt, a history of bipolar disorder, having less than a high school education, and the interaction between PHQ-8 score and a preinjury suicide attempt. With each 1-point increase on the PHQ-8, participants' risk of endorsing SI increased by 12% for those without a history of prior suicide attempts and by 2% for those with a history.

DISCUSSION

In the current sample, 25% of participants reported SI at some time during the first year after TBI, a rate that exceeds the general population by almost 7 times.³² This finding aligns with several previous studies that examined SI after TBI.^{11–13} Compared with the study by Tsousides et al. of patients in the chronic phase of their TBI recovery,¹² our study found a remarkably similar rate of SI (27.3% vs 24.7%). This is interesting given that their study included community-based participants whose history of TBI was based on self-report

and whose average time since injury was 5.93 years, whereas our study included a representative sample of hospitalized patients with medically confirmed TBI. In addition, our participants were followed over the first year after injury, during which time the point prevalence of SI remained fairly consistent. In a 2007 review of the literature, Simpson and Tate³³ concluded that there is not a circumscribed window of risk for suicidality after TBI. In contrast to suicide risk among individuals diagnosed with cancer, which is elevated during the first 5 years after diagnosis and then declines,³⁴ SI appears to increase after TBI and may remain chronically elevated.

Histories of premorbid psychiatric hospitalization, mental health diagnoses, and suicide attempts were more common among this sample than among the general population. Twelve percent of our sample reported at least 1 suicide attempt prior to their TBI, whereas lifetime rates in the general population range from 1.9% to 8.7%.³⁵ The majority of TBIs were unintentional, although 1.4% were the result of a suicide attempt, a finding that mirrors rates found in the TBI Model Systems Database.³⁶ Together, these findings align with the conclusion of Fann et al. that psychiatric

illness may increase one's risk of TBI.³⁷ Of those participants who denied a history of prior suicide attempts (n = 485), 20% reported SI in the first year after injury. The 12-month prevalence of SI among the general population ranges from 2.1% to 10.0%,³⁵ which suggests that even among patients who do not have a prior history of suicidal behavior, rates of SI are at least double those of the general population. These findings suggest that although mental illness may be overrepresented among individuals who sustain a head injury, TBI may independently increase the risk of SI.

When only those variables that are available via medical record review were examined, multivariate analyses demonstrated that being female, having less than a high school education, and having Medicaid were independent predictors of SI. Together, those variables accounted for 6.4% of the variability in SI, compared with the full model, which accounted for 25.5%. Multivariate analyses revealed that the independent predictors of SI during the first year after injury were initial depressive symptoms, preinjury psychiatric history (specifically, a diagnosis of bipolar disorder) and a prior suicide attempt. Consistent with studies in the general population,^{35,38} having less than a high school education was also associated with increased risk of SI. In the current sample, SI in people without a high school education was 10 times greater than would be expected among similarly educated individuals in the general population.³² Given that TBI may heighten financial stress and complicate return to work,³⁹ it is possible that TBI is associated with greater adjustment difficulties among individuals who may have fewer options or resources because of limited education. Although the sequelae of TBI may yield unique vulnerabilities among brain injury survivors, risk factors for SI among individuals with TBI appear to align with those that have been identified among the general population.

Future Research

The body of research investigating the mental health sequelae of TBI has grown immensely in recent years; however, there continue to be notable gaps in the literature. Further research is necessary to establish the prevalence of SI and behavior among TBI survivors.¹⁵ Likewise, additional investigation

TABLE 3—History of Preinjury Suicidal Behavior, Traumatic Brain Injury (TBI), and Psychiatric Diagnoses, by Presence or Absence of Suicidal Ideation (SI): Seattle, WA, June 2001–March 2005

Variable	SI Negative (n = 421), No. (%)	SI Positive (n = 138), No. (%)	SI Positive, % ^a	OR (95% CI)
History of suicide attempt(s)*				
No (Ref)	386 (93)	99 (73)	20	1.00
Yes	30 (7)	37 (27)	55	4.79 (2.73, 8.47)
History of psychiatric hospitalization*				
No (Ref)	395 (95)	117 (85)	23	1.00
Yes	21 (5)	20 (15)	49	3.21 (1.59, 6.46)
History of hospitalization for prior TBI				
No (Ref)	374 (90)	120 (87)	24	1.00
Yes	40 (10)	18 (13)	31	1.40 (0.73, 2.61)
Depression history*				
No (Ref)	338 (85)	81 (60)	19	1.00
Yes	61 (15)	53 (40)	46	3.61 (2.27, 5.76)
Bipolar or manic depression history*				
No (Ref)	407 (98)	112 (82)	22	1.00
Yes	9 (2)	24 (18)	73	9.64 (4.18, 24.26)
PTSD history*				
No (Ref)	391 (95)	116 (89)	23	1.00
Yes	22 (5)	14 (11)	39	2.14 (0.98, 4.54)
Other anxiety disorder history ^b *				
No (Ref)	397 (95)	116 (85)	23	1.00
Yes	19 (5)	20 (15)	51	3.59 (1.75, 7.38)
Lifetime alcohol dependence*				
CAGE ^c score 0–1 (Ref)	215 (62)	62 (51)	22	1.00
CAGE score ≥ 2	130 (38)	60 (49)	32	1.60 (1.03, 2.48)

Note. CI = confidence interval; OR = odds ratio; PTSD = posttraumatic stress disorder.

^aPercentage with SI (response ≥ 1 on Patient Health Questionnaire [PHQ-9] item 9) in each variable category.

^bOther anxiety disorders included generalized anxiety disorder, panic disorder, obsessive compulsive disorder, and any phobia.

^cThe CAGE (cut down, annoyed, guilty, eye opener) screening tool.²⁴

* $P < .05$; P values were determined by the Fisher exact test.

to delineate injury factors (e.g., injury severity) and postinjury factors (e.g., restriction of lethal means, provision of social support, return-to-work status) that may exacerbate or mitigate risk of suicide is warranted. At present, intervention studies for depression after TBI are few, and evidence-based treatment guidelines have not been established.⁴⁰ Developing interventions to address difficulties with depression among brain injury survivors should be a priority, and those interventions should be tailored to meet the challenges of access to care (e.g., transportation issues) and concomitant difficulties often present following TBI (e.g., cognitive difficulties). Future studies should

also be mindful in selecting appropriate comparison groups. As we are able to better understand factors associated with self-directed violence among individuals who have suffered a TBI, we will be more capable of preventing lives from being unnecessarily lost to suicide.

Clinical Implications

This study identified several predictors of SI that are extractable from the medical record. This information is valuable as it may assist providers in identifying very early in the inpatient stay which patients may be at increased risk for SI later in their recovery. The most potent predictors, however, were those that

related to depressive symptoms soon after injury and preinjury psychiatric history. The challenges that arise in the wake of a brain injury (e.g., concomitant injuries, financial concerns, return-to-work issues) are many, and attending to psychiatric well-being along with competing difficulties is important for optimizing outcomes. Our findings suggest that individuals who have serious psychiatric difficulties before sustaining a head injury are likely to continue to be vulnerable to suicidality (perhaps even more so) after injury. Assessing prior psychiatric history should be a routine component of health assessment. Our findings also underscore the importance of assessing mood during the early stages after injury, as this period may serve as a window of opportunity to identify patients who are at particular risk for suicidal behavior later in recovery. Previous research has suggested that risk for suicide remains elevated for at least the first 15 years after injury.⁵ TBI may therefore be viewed as a chronic health condition,⁴¹ and SI should be monitored long after injury. A measure of depression integrated into TBI care would be a useful component of mood assessment and would be feasible in many health care settings, including those that provide acute care. Ensuring that health care providers are sufficiently skilled in suicide assessment is also critical. Prior research has suggested that training in suicide assessment and management

TABLE 4—Multivariate Model Predicting Suicidal Ideation During the First Year After Traumatic Brain Injury, Using Variables Available in the Medical Record: Seattle, WA, June 2001–March 2005

Variable	OR (95% CI)	P
Male	0.59 (0.37, 0.92)	.021
Completed high school	0.47 (0.25, 0.88)	.017
Insurance		.007
Medicaid vs commercial or private	1.72 (1.09, 2.72)	.019
Medicare vs commercial or private	0.60 (0.29, 1.17)	.149

Note. CI = confidence interval; OR = odds ratio. Nagelkerke $R^2 = 0.06$.

TABLE 5—Full Multivariate Model Predicting Suicidal Ideation During the First Year After Traumatic Brain Injury: Seattle, WA, June 2001–March 2005

Variable	OR (95% CI) ^a	P ^a
Completed high school	0.37 (0.20, 0.71)	.002
First PHQ-8 score	1.12 (1.08, 1.16)	< .001
History of bipolar disorder or manic depression	4.60 (1.91, 11.06)	.001
Suicide attempt history	8.55 (3.06, 23.84)	< .001
PHQ-8 × suicide attempt history	0.91 (0.84, 0.99)	.026

Note. CI = confidence interval; OR = odds ratio; PHQ-8 = Patient Health Questionnaire (PHQ-9) with item 9 excluded.

Nagelkerke $R^2 = 0.26$;

^aAsymptotic estimates.

for health and mental health providers is inadequate^{42–45}; however, a recent training program implemented in Australia among rehabilitation professionals showed sustained improvements in knowledge and skills in suicide assessment at 6-month follow-up.⁴⁶ This topic merits further scrutiny.

Limitations

This study has several limitations that warrant discussion. First, we used a single, self-report question from the PHQ-9 to classify individuals as having SI. Thus, it relied on participants' interpretation of the question, which does not distinguish explicitly passive thoughts of death from active suicidality. In spite of this limitation, item 9 of the PHQ-9 has been found to have adequate specificity (0.84) and sensitivity (0.69) compared with interview measures that target suicidality specifically,⁴⁷ although some studies have suggested that this question overestimates rates of SI.⁴⁸ We also dichotomized responses on item 9 for the purposes of statistical analyses; thus, relationships presented do not reflect how frequently (i.e., several days, more than half of days, or nearly every day in the preceding 2 weeks) participants experienced SI at each assessment point. SI-positive participants also completed fewer interviews than SI-negative participants, which may have created conservative bias in the SI prevalence rates at each time point. Individuals who reported suicidality were referred for further assessment and intervention. It is possible that this had an impact on the persistence of SI.

This sample, although representative of the brain-injured patient population served by a large level 1 trauma center in the Pacific

Northwest, included an overrepresentation of Medicaid recipients and a relatively small number of patients from diverse racial/ethnic backgrounds. Thus, these findings may not generalize to regions that have a vastly different socioeconomic or cultural composition. Finally, we do not have longitudinal data on suicide attempts or completions that may have occurred during the study or after the project concluded.

Conclusions

This study adds to the burgeoning body of literature investigating SI after TBI by identifying risk factors for SI in a representative sample of adults within the first year after TBI. Identifying key risk factors for SI among patients with a recent TBI may help health care providers to determine who among their recently injured patients may be particularly vulnerable to SI during the year after TBI. Our findings suggest that symptoms of depression soon after injury, prior suicide attempt(s), a diagnosis of bipolar disorder, and low educational attainment are particularly salient risk factors. ■

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J. L. Mackelprang, C. H. Bombardier, J. R. Fann, and J. K. Barber were responsible for drafting the article. J. L. Mackelprang, C. H. Bombardier, N. R. Temkin, and J. K. Barber were responsible for statistical analysis and interpretation of data. C. H. Bombardier, J. R. Fann, N. R. Temkin, and S. S. Dikmen obtained funding to conduct the original parent study and were responsible for acquisition of the data. N. R. Temkin and S. S. Dikmen provided critical revision of the article for important intellectual content. All authors were responsible for conceptualization and design of the study.

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Human Participant Protection

This study was approved by the University of Washington Human Subjects Division.

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